The Red Scarcity: Do High Percentages of Young Adult Residents in California Counties Associate with Low Republican Voter Share?*

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This project examines whether the population proportion of young adults (ages 18–34) in California counties is negatively associated with Republican voter registration. Using 2025 county-level data, we fit a simple linear regression model with young-adult population share as the predictor and Republican voter share as the response. We find that the estimated slope parameter of our model was negative, suggesting that there is a negative association between counties young-adult population share and their Republican voter share. These findings provide motivation for further research into the potential relationship between age and party preferences.

Introduction

Understanding how key demographics of a population coincide with voter registration can help us understand the core values of their community, and the kinds of policies they may be likely to vote for. Nationwide trends over the last decade have shown that younger voters tend to associate in higher numbers with the Democratic Party compared to the Republican Party (Pew Research Center 2024). In this paper we zoom in and take a more local look at these trends inside California counties. We will see how the population of young adults in a county associates with the county's Republican voter share. Using 2025 U.S. Census Bureau data, and Voter Registration Statistics from the California Secretary of State, we aim to answer the question: Does a high proportion of young adults in a California county negatively associate with Republican voter share?

We investigate this question using a simple linear regression model that uses a county's young adult population proportion as a predictor, and their Republican voter share as our outcome.

*Project repository available at: https://github.com/TypeCharStar/M261A-Project-1.

We then perform inference on the slope of this model to test for a statistically significant negative association between the variables. After analyzing our model and conducting inference, we find that there is statistically significant evidence of a negative association between young adult population share and Republican voter share in California counties. This relationship is important because it could prompt further research into why young adults in California are choosing to politically affiliate less with the Republican party.

The rest of the paper first takes a look at where our data was sourced from and how it was compiled and transformed for this paper. Then we will look at the methods used, specifically the assumptions and definition of our model. After that we present the results of our model and subsequent inference performed. Finally, we will discuss our analysis conclusions, and address the limitations of our study, as well as potential areas for further research.

Data

To perform my analysis, I combined datasets from the U.S. Census Bureau - 2025 County Population by Characteristics (U.S. Census Bureau 2026a) and California Secretary of State - 2025 Report of Registration.(California Secretary of State 2025).

Our observational units are all 58 California Counties. The variables used in our analysis include: `county`, which contains the name of each county, `ya_pop_pct`, our predictor variable, which is the percentage of 18-34 year old residents in county, and `reg_rep_oo_elig`, our response variable, which is the percentage of registered Republican voters in a county. Both variables are measured using the most recent full year data, which is for 2025. Since we plan to fit a simple linear regression model, we limit our observations to a single year. This aids in preventing any correlation of errors that can arise from time-series data. Using the **most recent** year also allows us to focus on current political trends.

The US Census Bureau estimates a county's population every year after the decennial census using a demographic balancing equation that starts with a base estimate of the county's previous year population, then adds the difference between births and deaths in the county. Then migration into and out of the county is accounted for by analyzing data from the IRS, Medicare enrollment, SSA Identification files, and the change in the Group Quarters population. This produces a new population estimate for a given year. This estimation procedure yielded a mean absolute percent error of only about 2.9% across all counties' population estimates between the 2010 and the 2020 Census. More details on this process can be found in methodology statement for the 2025 estimates (U.S. Census Bureau 2026b).

Voter registration data is tracked by the California Secretary of State office. They do not have any documentation on exactly how they compile these numbers, but they note that the data is obtained from all sources of voter registration such as the DMV, by Mail, State Agencies, Armed Forces registration, etc. Duplicate registrations and deletions from the voter roll are also tracked. More information on how the number of registered voters is calculated can be

found on their webpage regarding federal reporting to the ERC (California Secretary of State, n.d.).

We chose the age range 18-34 to measure young adult population since it is a range the U.S. Census Bureau has used in recent years to broadly define the term (Vespa 2017). Our predictor variable was calculated by dividing the number of residents aged 18-34 in a county by the total number of residents in that county.

We chose to measure our response variable, the proportion of Republican voters in a county, by dividing the number of registered Republican voters by the total number of people eligible to vote in a given county. We used the number of eligible voters instead of total county population to control for people who live in the county but are ineligible to register to vote, for a variety of reasons (too young, not a citizen, felon, etc.)

Below, we view some descriptive statistics about our predictor and response variables.

Table 1: Descriptive Statistics for Young Adult Population Percentage and Republican Voter Share of California Counties

	N	Mean	SD	Minimum	Median	Maximum
Young Adult Pop Pct	58	21.543	4.290	11.136	22.300	34.817
Reg Republican / Eligible Voters	58	27.475	9.904	6.179	27.287	46.588

We can interpret the unit of measurement for Young Adult Population Percentage as the number of young adults per 100 residents in a California county. For Registered Republicans out of Eligible Voters our unit of measurement can be interpreted as the number of registered republican voters per 100 residents eligible to vote in a given county.

We notice that the mean population percentage of young adults in a county is 21.543%. The county with the lowest population percentage of young adults is Sierra county, where young adults only make up 11.136% of the population. In contrast, Yolo county has the highest percentage of young adults in their population, at 34.817%.

When viewing the percentage of eligible voters registered as Republican in a county, the statewide mean is 27.475%. San Francisco county comes in with the lowest percentage of registered Republican voters, at only 6.179%. On the other hand, Amador county has the largest percentage of registered Republican voters at 46.588%.

The mean is within 1 percentage point of the median for both variables. This tells us that the distribution of both variables is close to symmetric. We also do not observe any extreme values; the spread of observations for both variables stays contained within 3.1 standard deviations of their respective means.

Methods

To model the relationship between our variables, we utilize a simple linear regression model with population percentage of young adults as the predictor, and percentage of registered Republican voters as our response. For:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Y_i = The percentage of registered Republican voters in the i_{th} county.
X_i = The population percentage of young adults (Age 18-34) living in the i_{th} county.
β_0 = Our intercept parameter. This can be interpreted as the registered Republican voter share of a county if no young adults reside there.
β_1 = Our slope parameter. This can be interpreted as the percentage-point change in registered Republican voter share per percentage-point increase in the population share of young adults in that county.
ε_i = Uncertainty in the Republican voter share of the i_{th} county that can not be explained by our predictor variable.

We fit our model using the `lm()` function in R (R Core Team 2026). All plots are created using the `ggplot2` library.

There are a hierarchy of assumptions that must be met for our simple linear regression model to be valid, including normality of errors since we plan to conduct inference on our model.

Assumptions:

1. Linearity Our mean function, $E(Y_i|X_i) = \beta_0 + \beta_1 X_i$, must be correctly specified.

When viewing a scatterplot of our data, we see that points follow a vaguely linear shape. Furthermore, when viewing a residual vs predictor plot, we see a healthy scatter of points around the the $y = 0$ line, which supports our assumption that the data is linearly related.

2. Independence of Errors For all $i \neq j$, it must be the case that $E(\varepsilon_i|\varepsilon_j) = E(\varepsilon_i)$. In other words, the value of one error term should have no impact on the value of another.

By limiting our data to a single year point, we control for any dependence of errors caused by time. However, we can not rule out the possibility that the location of counties may influence one another. For 2 bordering counties, it is possible that the error terms could be correlated. We will mind this potential violation of the model assumptions as we discuss our results.

3. Equal Variance of Errors All error terms are be identically distributed with a mean of 0, and variance σ^2.

Upon viewing a residual vs predictor plot for our model, we do not see any heteroskedasticity in the plot. The residuals for the predictors form an even scatter about zero, with no obvious clustering of residuals visible on the plot.

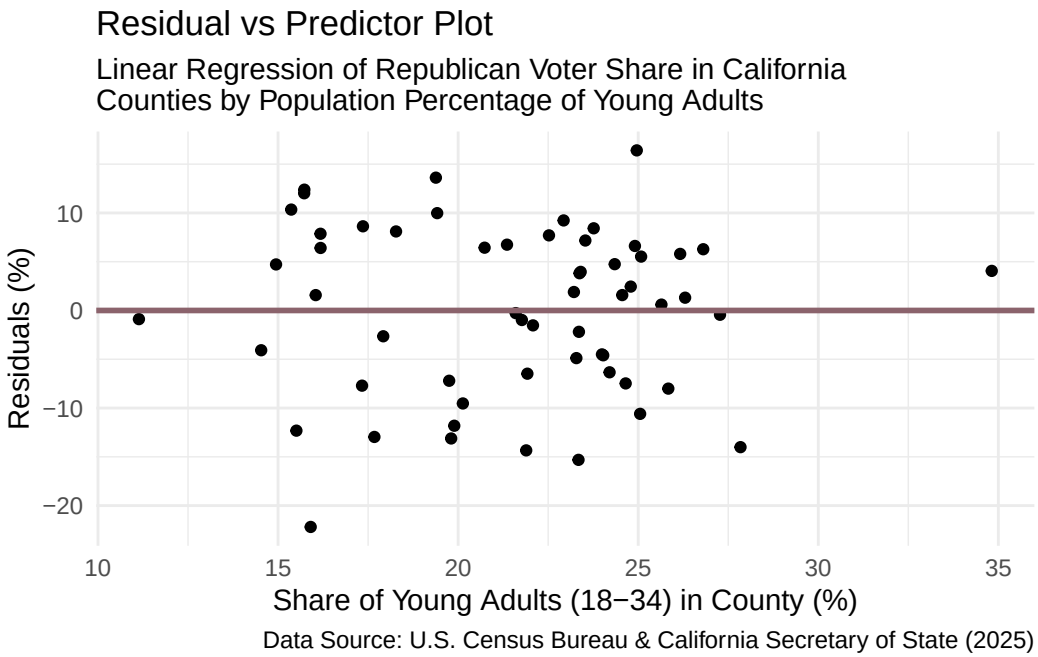


Figure 1: Residuals vs. predictor plot for the simple linear regression model: (Republican Voter Share ~ Young Adult Population Share), where each observation represents a California county. Points represent how our model's prediction differs from the actual Republican voter share of a county. We do not observe any heteroskedasticity in the plot, and see a healthy spread of observations around the 0 line.

4. Normality of Errors All error terms are identically distributed as $N(0, \sigma^2)$.

When we create a normal Q-Q plot of our residuals, we find that they cluster around the line of equivalence, with slight departures at the tail ends.

Since our sample size is reasonably large, at $n = 58$, the Central Limit Theorem can be applied to compensate for these deviations, letting us assume approximate normality. Due to this, our model as well as our inference tests should still be robust, allowing us to find statistically significant associations, as long as all other assumptions are met.

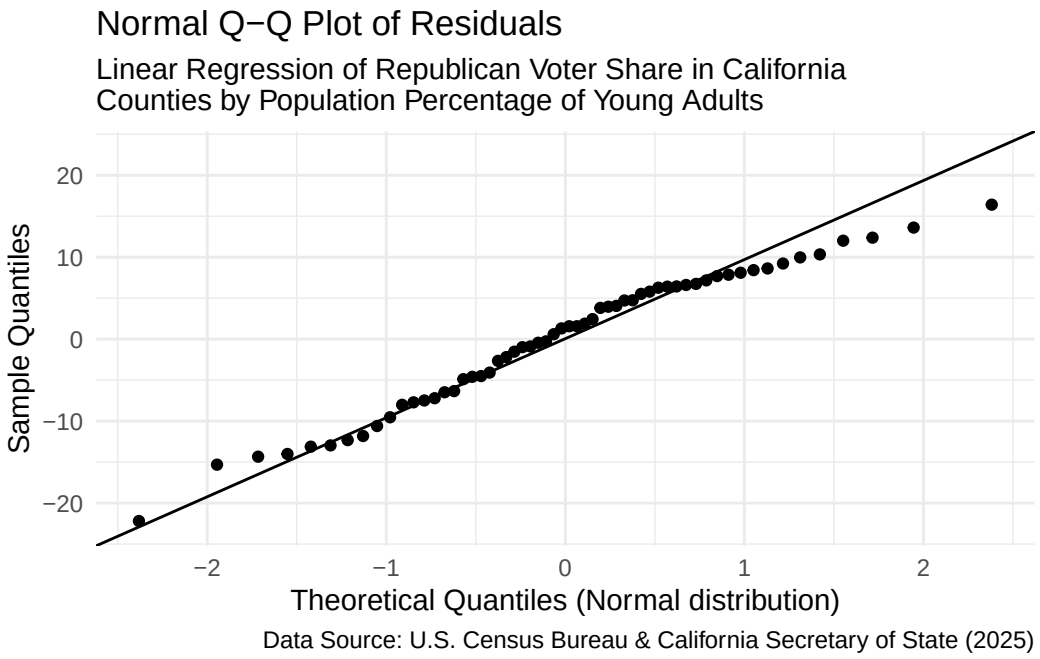


Figure 2: Normal Q–Q plot of model residuals. Points close to the qq line suggest approximate normality, while deviations from the qq line suggest departure from normality. We observe most residuals clustering around the qq line with some departure at both endpoints.

We note that our outcome variable represents a percentage and is thus bounded between 0 and 100. Bounded outcome variables can hinder a model's ability to meet the required assumptions of simple linear regression, especially when observations gather towards the endpoints of the bounds. In our case, we see that our observed Y_i's lie well within our endpoints, spanning from 6 to 47 percent. Since we do not observe large clusterings of observed outcomes near 0 or 100, we are not particularly concerned with possible non-linearity or issues with our model errors that would arise from a bounded response.

Results

Below we see the fitted simple linear regression line imposed on a scatter plot of our variables of interest. Our fitted model takes the equation:

$$\widehat{(\text{Repub Voter Share})} = 52.406 - 1.157(\text{Young Adult Pop Share})$$

Where $b_0 = \hat{\beta}_0 = 52.406$ and $b_1 = \hat{\beta}_1 = -1.157$. This negative slope parameter is of interest to our analysis. We can interpret the slope as: for each 1% increase in the proportion of young adult residents in a county, we expect the Republican voter share to decrease by 1.157%.

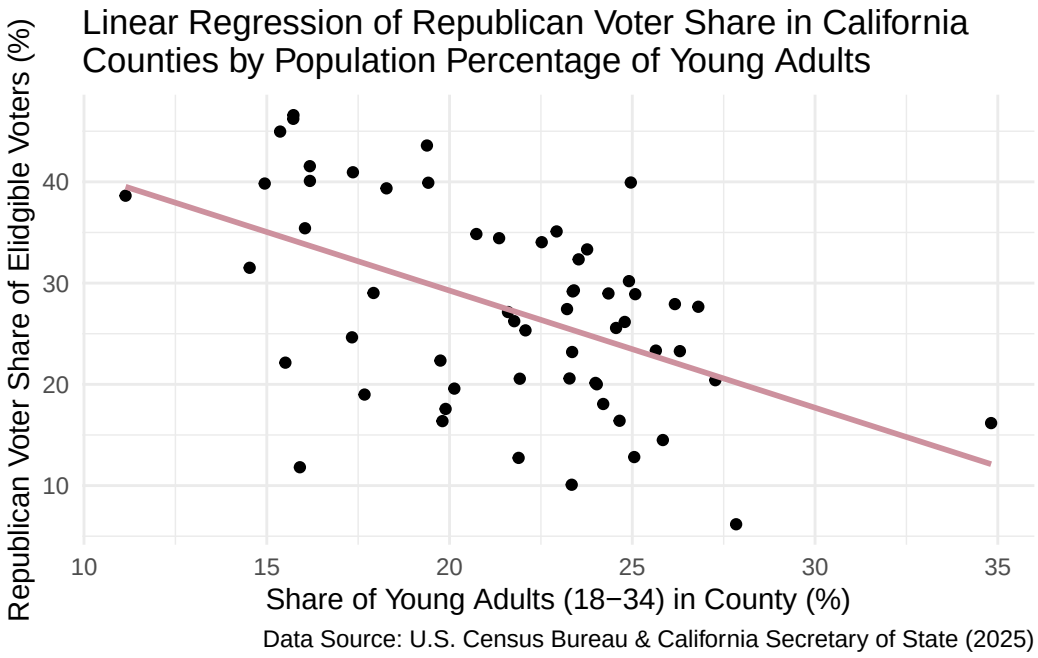


Figure 3: Simple Linear Regression model of (Republican Voter Share ~ Young Adult Population Share). Each 1 percent increase in Young Adult Population Share associates with a 1.157 percent decrease in Republican Voter Share. The R^2 value of this model is 0.251, implying 25% of the variation in data could be explained by our model.

The coefficient of determination R^2 measures the the proportion of variation in the data explained by our regression model. We observe $R^2 = 0.251$. This is a low to moderate value, but we find it acceptable since there is a lot of uncertainty when creating models based on human behaviors. We do note that this could mean there are other factors with considerable influence over Republican voter share left out of our model.

To test if there exists a statistically significant negative association between the share of young adults in a county, and Republican voter share, We conduct a one-sided t-test with an $\alpha = 0.05$ Type I error rate on β_1. Our Hypotheses are:

$$H_0 : \beta_1 = 0 \qquad \text{vs} \qquad H_1 : \beta_1 < 0$$

Our test statistic is:

$$t^* = \frac{b_1}{\widehat{se(b_1)}} = \frac{-1.157}{0.267} = -4.\overline{3}$$

We can calculate the p-value of t^* as:

$$p = P(T_{56} \leq t^*) = 3.057212 \times 10^{-5}$$

Our p-value is much much smaller than our 0.05 Type I error rate. We can conclude that there is evidence of a statistically significant negative association between the share of young adults in a county, and Republican voter share.

Discussion

Through our analysis, we confirmed that there exists statistically significant evidence of a negative association between the percentage of young adults in California counties, and the Republican voter share. Our fitted model resulted in a slope parameter that can be interpreted as: for each 1% increase in the proportion of young adult residents in a county, we expect the Republican voter share to decrease by 1.157%.

Showing that a negative association exists between our variables is interesting and could lead to broader conversations about how political values differ by generation in California. We could use evidence of this generational difference as a launching point to explore what social and environmental factors could be prompting young people in California to affiliate less with the Republican party compared to older residents. It would also be interesting to see if this association persists in other states, the country as a whole, or even in other countries with similar right-wing parties.

Our study is not without its limitations. Firstly, we can not rule out the possibility that some error terms are dependent based on location. This dependence could make the results of our t-test appear more statistically significant than they actually are. A potential improvement of this study could include location-independent samples of counties from regions all over the country. So we could broaden our research question, and to try and control for this. Secondly, we observed a relatively low R^2 value. While human behavior is difficult to model perfectly, it would be interesting to see if a multiple regression model, where we could include other demographic variables, would yield a model that explains more of the variation.

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